Some topics in decision theory
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1. A note on 'small world' versus 'big world' decision problems

On pretty much any philosophically viable interpretation of 'utility', the utility value assigned to a state-of-affairs will have to be sensitive, in a quite fine-grained way, to the details of what's going on there. Even if we understand utility as a measure of the strength of the agent's preferences, and our agent is monomaniacally fixated on having as high a bank balance as possible at some particular time, still the value of this will often depend on many factors beyond those represented in the simple decision matrices that get used in examples.

But one nice feature of EU theory (which distinguishes it from some competitors) is that it lets us ignore a whole lot of our uncertainty about utility so long as we don't take it to be correlated with our current action.

For example: suppose that in addition to being uncertain where the roulette ball will land, you're also uncertain whether (D) your rich uncle will die and bequeath you $1000. So your *real* decision matrix is this:

	Red∧D	Black∧D	Green∧D	Red∧¬D	Black∧¬D	Green∧¬D
Bet all on red	$1,100	$900	$900	$100	-$100	-$100
Bet all on black	$900	$1,100	$900	-$100	$100	-$100
Don't bet	$1,000	$1,000	$1,000	$0	$0	$0

For each act A, we can break up $E_{Pr}(U(A))$ as $E_{Pr(\cdot|D)}(U(A))Pr(D) + E_{Pr(\cdot|\neg D)}(U(A))Pr(\neg D)$,. But for each act A, $Exp_{Pr(\cdot|\neg D)}(U(A))$ is the same as $Exp_{Pr}(U(A))$ in the old, simpler decision problem, while $Exp_{Pr(\cdot|\neg D)}(U(A))$ is this number plus $1000. So the *real* EU of each act is just its EU in the simple problem + $1000 Pr(D): it's thus harmless to use the simple representation when you're deciding how to bet.

- So long as you value dollars linearly, this goes through even if you treat your uncle's death as more likely given red than given black!

Beware that alternatives like 'risk-weighted expected utility theory' disrupt this simplification.

2. What are the "states"?

Philosophers (like me) who are comfortable with 'proposition'-talk can understand the "states" for a given decision problem as *propositions*. We might require them to always be chosen so as to be pairwise incompatible and jointly necessary; or require that the proposition that that more than one of them is true or that none of them is true have zero probability (in whatever we take to be the operative sense of 'probability'); or at least require that the possibility of this not being the case be somehow negligible.

To get non-absurd results, it is crucial that the "states" should be propositions whose truth or falsity is (in some sense that remains to be clarified) *entirely beyond the agent's control*, at the time when the decision is being made The following example due to Joyce (1999) illustrates what goes wrong if we don't impose this:

	Windshield broken	Windshield not broken
Pay protection	–$510	-$10
Refuse to pay	–$500	$0

There are two dominant philosophical approaches to cashing out this notion of 'entirely beyond the agent's control'. The tradition of "causal decision theory" (Stalnaker, Gibbard, Harper, Lewis, Skyrms, Eells…) brings in concepts like *causation*, *counterfactuals*, and/or *chance.* By contrast, according to "evidential decision theory", it's sufficient that the states be *probabilistically independent* of the acts (according to the agent's credences).

The contrast between these schools is brought out by *Newcomb's Problem.* Causal decision theorists favor the following representation (where two-boxing dominates):

	$1m in opaque box	Nothing in opaque box
Take both	1,001,000	1,000
Take one	1,000,000	0

Causal decision theory is motivated in the first instance by the intuitive judgment that one ought to take both boxes in this case, and analogous judgments about other cases (though of course one can seek other, more theoretical arguments).

By contrast, evidential decision theorists will have no problem with the following representation, where one-boxing maximizes EU so long as one has high enough credence that the predictor got it right:

	Predictor accurate	Predicator inaccurate
Take both	1,000	1,001,000
Take one	1,000,000	0

Note that when each state is probabilistically independent of each act, the formula for EU I gave last week, namely $\mathrm{EU}(a) = \sum_s U(a \wedge s)\mathrm{P}(s)$, can be restated as $\sum_s U(a \wedge s)\mathrm{P}(s \mid a)$. Assuming there are only finitely many possible utility levels and letting $U{=}u$ be the proposition that the value of U is x, this is equivalent to defining $\mathrm{EU}(a)$ as $\sum_u u\mathrm{P}(U{=}u \mid a)$—the expectation of u conditional on a—a formula in which we don't need to use the concept of a *state* at all.

3. Some tricky cases for the Savage formalism

There are some very straightforward gambling cases that are, surprisingly, tricky to model in Savage's framework—for example, the decision whether to take a bet that costs \$1 and pays off \$3 if a certain coin lands heads, where the coin won't be tossed unless you take the bet. It seems absurd to have propositions like 'the coin lands heads' as states: how should we fill in the utility value of the cell corresponding to the combination of this state with the action *decline the bet*?

The evidential decision theorist has no trouble modelling such cases, since they don't need the concept of a state.

Some causal decision theorists propose the following generalization of Savage's formula which does not require that the combination of any act and state pin down a unique utility value:

$$\mathrm{EU}(a) = \sum_s \sum_{\mathrm{u}} u\mathrm{P}(U{=}u \mid a \wedge s)\mathrm{P}(s) = \sum_s \mathrm{E}_{\mathrm{P}(\cdot \mid a \wedge s)}(U)\mathrm{P}(s)$$

A similar thought: suppose that we identify *state-propositions* with propositions of the form Ch=Q, that fully specify a certain probability function Q as the *chance* function at the time of decision. Then we could use these chance functions to take expectations within each state:

$$\mathrm{EU}(a) = \sum_Q \sum_{\mathrm{u}} u\ \mathrm{Q}(U{=}u \mid a)\mathrm{P}(Ch{=}Q) = \sum_Q \mathrm{E}_{Q(\cdot \mid a)}(U)\mathrm{P}(Ch{=}Q)$$

This will coincide with the previous proposal if we take it that rational agents ought to obey the Principal Principle, having $\mathrm{P}(b \mid Ch{=}Q) = Q(b)$.

A different way of filling things in (Stalnaker, Gibbard, Harper) appeals to *counterfactuals* to find state-propositions that can be plugged in to a Savage style representation of the case:

	Coin would land heads if tossed 0.5	Coin would land tails if tossed 0.5
Decline bet	0	0
Take bet	2	–1

This approach only yields sensible results if one takes counterfactuals to satisfy the principle of *Conditional Excluded Middle*, $p{>}q \vee p{>}\neg q$. This is intuitive, but raises metaphysical puzzles—it's hard to understand what makes the difference between worlds where the coin isn't tossed and it would have landed heads if tossed and worlds where the coin isn't tossed and it would have landed tails if tossed.

So long as you are OK with counterfactuals that satisfy CEM, it seems harmless to equate "state propositions" with conjunctions of the form $a_1{>}U{=}u_1 \wedge \ldots \wedge a_n{>}U{=}u_n$ — conjunctions that say, for each available action, what the utility of the world *would* be if that action were performed.

If one makes this assimilation, the Savage-style formula $\mathrm{EU}(a) = \sum_s U(a \wedge s)\mathrm{P}(s)$ becomes equivalent to $\mathrm{EU}(a) = \sum_u u\mathrm{P}(a{>}U{=}u)$: the expectation (according to one's credence function P) of a random variable whose value at a given world w is the value of U at the world w' such that, at w ($a > w'$ is actualized). (According to a popular metaphor: the 'closest' world to w where action a is performed.)

Counterfactual decision theory will agree with chance decision theory so long as (i) the agents obey the Principal Principle, and (ii) the agent is certainthe chances at the relevant time obey "Skyrms's principle": $\mathrm{Ch}(U{=}u \mid a) = \mathrm{Ch}(a{>}U{=}u)$. For then, we'll have

$$
\begin{aligned}
\mathrm{EU}(a) &= \textstyle\sum_Q \sum_u u\ Q(U{=}u \mid a)\mathrm{P}(Ch{=}Q) \\
&= \textstyle\sum_Q \sum_u u\ Q(a{>}U{=}u)\mathrm{P}(Ch{=}Q) \text{ (by Skyrms's principle)} \\
&= \textstyle\sum_Q \sum_u u\ \mathrm{P}(a{>}U{=}u \mid Ch{=}Q)\mathrm{P}(Ch{=}Q) \text{ (by the Principal Principle)} \\
&= \textstyle\sum_u u\ \mathrm{P}(a{>}U{=}u) \text{ (by the Principal Principle)}
\end{aligned}
$$

4. Allais's paradox, and some alternatives to EU theory

EU theory (in Savage's formalism) validates the "sure thing principle": if acts a and b differ only on states in a certain proposition p and a is weakly preferable to b, and acts a' and b' differ only on states in p and agree with a and b respectively on those states, then a' is weakly preferable to b'.

A famous example due to Allais has been taken to pose problems for the Sure Thing Principle (at least on a descriptive interpretation).

	Ball 1–89 drawn	Balls 90–99 drawn	Ball 100 drawn
a	1000	1000	1000
b	1000	5000	0
a′	0	1000	1000
b′	0	5000	0

Here the states specify which of 100 balls is drawn from a certain urn and the payoffs are in dollars. The empirically confirmed observation is that people seem to prefer a to b, but b′ to a. There is no way of converting utilities to dollars where this can be rationalized by EU maximization. If one takes this to be *rational* or normatively permissible, it poses a problem for normative EU theory.

- One possible defence on behalf of EU theory (understood in the usual way as a theory of rationality applicable to arbitrary utility functions) is that the relevant people assign lower utility to worlds where they get nothing and they had safe option where there was no chance of getting nothing than to worlds where they get nothing and had no such safe option. (Maybe the pangs of regret are worse in the former worlds?)

If you don't buy this kind of defence, how could we modify EU theory to permit both of the popular choices in the Allais cases?

One possibility is just to get more permissive. E.g. one could adopt the super permissive theory that all and only non-dominated actions are permissible. Or the somewhat less permissive theory that all and only actions that aren't *stochastically dominated* are permissible, where a stochastically dominates b iff $\mathrm{P}(U{\geq}u \mid a) \geq \mathrm{P}(U{\geq}u \mid b)$ for all u and $\mathrm{P}(U{\geq}u \mid a) > \mathrm{P}(U{\geq}u \mid b)$ for some u.

Such permissive views cry out for supplementation of some kind if we're looking for a theory of something along the lines of "rational action-dispositions". We might want to have some kind of negative evaluation of certain *packages* of action-dispositions even if we don't want to say that any *individual* action that these dispositions would produce is impermissible.

- The 'risk-weighted EU theory' recently defended by Lara Buchak is an example of what this might look like. According to it, a rational person has a specific 'risk function' r: $[0,1]\rightarrow[0,1]$ (required to be monotonically increasing), and always chooses actions that maximise a certain quantity called 'risk weighted EU' that uses r to boost (or penalize) the contributions coming from the worst outcomes. For every option that isn't stochastically dominated we can find *some* r where it maximizes r-weighted

EU. But so long as one sticks to the same *r*, one will avoid certain kinds of egregiously exploitable patterns of behavior across time.

5. Savage's representation theorem

Savage proved a deep *representation theorem* that feels like it should be highly relevant to the project of defending EU theory (under various interpretations), as well as the project of explaining its truth assuming one takes it to be true.

To state the theorem, one needs a very abstract notion of "act" where *every* function from "states" to "outcomes" corresponds to an "act". The theorem says that for every "weak preference" orderings $\geq$ on this big set of acts that meets certain conditions, there exists a unique probability function P (defined on sets of states) and a utility function U (defined on outcomes) such that $f \geq g$ exactly when $E_P(U \circ f) \geq E_P(U \circ g)$. Moreover the conditions in question are satisfied by every $\preccurlyeq$ generated in this way from a U and a P, so long as U is nontrivial.

Here are the seven axioms (stolen from a handout of Patrick's—thanks!)

Axiom 1. $\geq$ is reflexive, transitive, and complete.

Definition A: h^f_A is the function that agrees with f on states A and h on states not in A

Axiom 2 (Sure Thing Principle): $h^f_A \geq h^g_A$ iff $j_A^f \geq j_A^g$.

Definition B: $f \geq_A g$ iff $h^f_A \geq h^g_A$ for any (or equivalently, for some) act h.

Definition C: A is *null* iff $f \sim_A g$ for any acts f, g.

Axiom 3. $x \geq y$ iff $f^x_A \geq f^y_A$ whenever A is non-null and x and y are *constant* acts.

Axiom 4. $b^a_A \geq b^a_B$ iff $d^c_A \geq d^c_B$, whenever a,b,c,d are constant acts such that $a>b$ and $c>d$.

Definition D: $A \geq B$ iff $b^a_A \geq b^a_B$ for some constant acts $a>b$

Axiom 5. $f > g$ for some f, g.

Axiom 6. Whenever $f > g$, there exists a partition, $\{A_1, ..., A_n\}$ of the set of states such that for any i: $f^h_{Ai} > g$ and $f > g^h_{Ai}$.

Axiom 7. If $f \geq_A x$ for every constant act x that coincides with g on some state, then $f \geq_A g$, and similarly for $\leq$.

6. The normative significance of Savage's theorem